

choice for the server, and each sensor node stores only an embedded key such that a compromising/captured node cannot reveal much security information of the sensor network.

Public-key-cryptography-based schemes. Public-key cryptography is considered very expensive for small sensor nodes, because typical public-key algorithms, e.g., RSA, require extensive computations and are not suitable for tiny sensors. However, the recent implementation of 160-bit elliptic curve cryptography (ECC) on Atmel ATmega128, a CPU of 8 MHz and 8 bits, demonstrates that ECC public-key cryptography is feasible for sensor nodes. Compared to symmetric key cryptography, public-key cryptography provides a more flexible and simpler interface, requiring no key predistribution, no pairwise key sharing and no complicated one-way keychain scheme.

Time synchronisation. Due to the collaborative nature of sensor nodes, time synchronisation is very important for many sensor network operations, such as coordinated sensing tasks, sensor scheduling (sleep and wake), mobile object tracking, time-division multiple-access (TDMA) medium access control, data aggregation and multicast source authentication protocol. For example, in the target tracking application shown in Fig. 1, sensor nodes need to know both the location where and time when the target is sensed in order to correctly determine the target's moving direction and speed. The network time protocol (NTP) is used for synchronisation in the Internet. A sensor network is a resource-constrained distributed system, and the NTP cannot be directly used by sensor networks.

Key management schemes

Some key management schemes for establishing and distributing cryptographic keys in the network are:

Distributed key scheme. A group of users of a network, referred to as a 'conference,' in order to securely communicate over public channels, could decide to use symmetric encryption

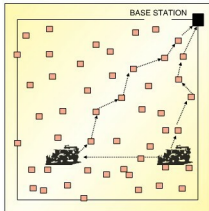


Fig. 1: Target tracking

algorithms, e.g., RC6 or AES. These algorithms are fast and presumed to be secure. However, to apply this strategy, they need a common key to encrypt and decrypt the messages they wish to send to each other.

Using a KDC is a common solution to the key establishment problem. A KDC is a server of the network which generates and distributes on-demand the conference keys. The idea is: each user shares a secure point-to-point channel with the centre. When one user wants to communicate with other users securely, he sends a request message for a session key. The centre checks for membership of the user in the group, and distributes the conference key to each member of the group in encrypted form.

A distributed key distribution centre (DKDC) is a set of 'n' servers of a network that jointly realise the function of a KDC, as shown in Fig. 2. In this setting, a user who needs to participate to a conference sends a key-request message to a fixed-size subset at his choice of the 'n' servers. The contacted servers answer with some information enabling the user to compute the conference key.

Assume every message between sensor nodes is encrypted and authenticated by the pairwise keys between them. The aim of distributed key establishment scheme is just to establish pairwise keys between sensor nodes.

Single network-wide key scheme. Using a single network-wide key is by far the simplest key establishment

technique. In the initialisation phase of this technique, a single key is preloaded into all the nodes of the network. After deployment, every node in the network can use this key to encrypt and decrypt messages. Only a single key is to be stored in the nodes' memory, and once deployed in the network, there is no need for a node to perform key discovery or key exchange since all the nodes in communication range can transfer messages using the key which they already share. This scheme counters several constraints with less computation and reduced memory use, but it fails in providing the basic requirements of a sensor network by making it easy for an adversary trying to attack.

Pairwise key establishment scheme. The pairwise key establishment scheme includes node-to-node authentication and resilience to node replication. For a network of 'n' nodes in the pairwise scheme, the key predistribution is done by assigning each node a unique pairwise key with all the other nodes in the network. With each node sharing a unique key with every other node in the network, this scheme offers node-to-node authentication. Each node can verify the identity of the node it is communicating with. This scheme also offers increased resilience to network capture as a compromised node does not reveal information about other nodes that are not directly communicating with the captured node.

One drawback with the pairwise scheme is the additional overhead needed for each node to establish n-1 unique keys with all the other nodes in the network and maintain those keys in its memory.

Public-key schemes RSA and ECC. Both RSA and ECC have been in research for many years. RSA stands for 'Rivest Shamir Adleman' algorithm. It was developed in 1977 and is still one of the most popular public-key encryption technologies currently available. RSA derives its strength from the complication of factoring very large numbers.

ECC was developed in 1985 independently by Koblitz and Miller. Its